Physical Chemistry CUME Feb. 13rd, 2015

Electron Paramagnetic Resonance (EPR) or Electron Spin Resonance (ESR)

EPR or ESR is a technique to study of radicals or transition metals.

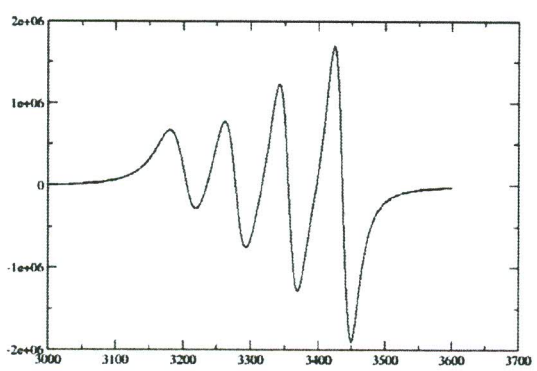
1. In a typical EPR experiment, a magnetic field about 3500 Gauss is used. In a magnetic field the energy of an electron with a spin quantum number m_s can be written as

$E = g_e \mu_B m_s B$, where $\mu_B = \frac{eh}{4\pi m}$, is called the Bohr magneton, which is 9.274×10^{-24} J/T and g_e is the gyromagnetic ratio, which is 2.0023 for the electron. Please work out the radiation frequency that is needed for a free electron to transit from the lower energy spin state to the higher energy spin state. (1Gauss=100μTesla, $h=6.626\times10^{-34}$ J·s)

2. The electron spin couples with the nuclear spin to cause level splitting. (fine structure).

The following EPR spectrum was obtained for a transition metal cation with a single unpaired d electron. What is the nuclear spin of the transition metal?

What are the possible nuclear spin values along the direction of the magnetic field?



This is actually a derivative spectrum but the number of peaks remains the same.

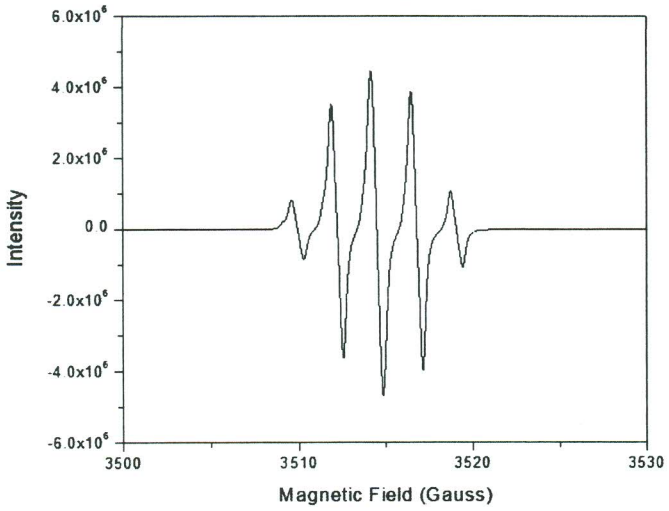
3. The EPR spectrum of the hydroquinone radical is shown below. Only 5 peaks are seen since the experiment is done in basic solution thus only the 4 protons on the ring couples to the unpaired electron. Each proton has a nuclear spin of ½.

Hydroquinone

What is the expected relative intensity of the 5 peaks ignoring the Boltzmann factor?

Assume the sum of nuclear spins along the magnetic field (Z) is S_z and the maximum possible value is S_z^{max}.

Show that it is 6 times as likely to have a S_z of zero than a S_z of S_Z^{max}.



4. The water molecule has 2 protons. If the EPR of a water molecule in the gas phase is measured, how many peaks do you anticipate to see?